

# Virinderpal Singh Batth

Lead Data Engineer

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## Professional Summary

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Data Engineering leader with 5+ years of experience designing and implementing enterprise-scale data solutions in financial services and insurance. Proven track record of driving significant performance optimizations, pioneering cloud and real-time streaming technologies, and leading cross-functional teams. Expert in Snowflake, AWS, Hadoop, and modern data stack technologies including dbt, Kafka, and PySpark.

## Experience

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### Lead Data Engineer

**evolv Consulting**

Oct 2025 – Present

- Leading a team of 4 data engineers to architect an insurance client's first unified operational data store in Snowflake using dbt
- Navigating complex ambiguities across legacy data platforms from multiple sub-companies to identify data overlaps and establish master data definitions
- Architecting high-performance data pipelines serving a transactional Kong API layer for real-time business insights

### Data Engineer

**USAA**

Apr 2021 – Oct 2025

- Resolved critical Snowflake performance bottleneck, reducing compute usage by **95%** and significantly decreasing query costs
- Optimized SCD Type 2 queries achieving **90% reduction** in running time (3 hours → 15 minutes) and 70% reduction in data scan volume
- Designed Kafka streaming pipeline for ML model training with staging tables, automated resiliency checks, and idempotent loading
- Led platform modernization migrating **2+ terabytes** from Hadoop to Snowflake using PySpark and parquet format
- Drove Secured Card to Credit Card transition resulting in **30% increase** in member engagement
- Implemented decoupled-push architecture reducing On-Call overhead by **90%**
- Architected cross-organizational data lake POC with AWS S3, reducing transfer time by 50%
- Enhanced PII/PCI/PHI pipeline security with data masking, tokenization, and RBAC

Software Engineer – ETL & Big Data Consultant

HCL America Inc.

Client: USAA | Nov 2017 – Apr 2021

- Pioneered the first project at USAA to utilize AWS cloud and real-time streaming (NiFi/Kafka), transforming batch to real-time processing
- Established AWS S3 to HDFS file transfer routes, enabling cloud-to-on-premise data integration
- Developed 30+ critical decisioning data pipelines in Hadoop supporting credit card risk decisioning
- Built core ETL pipelines using IBM DataStage and DB2 with complex transformations and governance checks
- Created automated schema check process eliminating hours of manual inspection
- Supervised Hadoop projects and mentored colleagues on CI/CD best practices
- Ensured PII/non-PII compliance with Information Governance standards

Technical Skills

| CATEGORY         | TECHNOLOGIES                                                          |
|------------------|-----------------------------------------------------------------------|
| Programming      | Python (PySpark, Pandas, Polars, SQLAlchemy, FastAPI), SQL, Bash, Git |
| Data Engineering | dbt, Kafka, Flink, NiFi, Hadoop, IBM DataStage                        |

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| Cloud & Platforms | AWS (S3, EC2, Redshift, Lambda, Athena), Snowflake, DB2  |
| Data Formats      | JSON, Parquet, REST APIs                                 |
| Visualization     | Apache Superset, Looker, Tableau                         |
| Practices         | CI/CD, Data Governance, RBAC, Data Masking, Tokenization |

## Education

**Bachelor of Science in Computer Science** Concentration: Big Data & Analytics New York Institute of Technology | *Cum Laude*

## Key Achievements

-  **First-Mover Innovation** – Delivered USAA’s first AWS cloud and real-time streaming project
-  **95% Compute Reduction** – Resolved critical Snowflake performance bottleneck
-  **90% Query Optimization** – Reduced SCD Type 2 queries from 3 hours to 15 minutes
-  **30% Engagement Lift** – Led data initiative driving measurable business outcomes
-  **2TB+ Migration** – Modernized platform from Hadoop to Snowflake
-  **90% On-Call Reduction** – Implemented automation reducing operational overhead